

WEST WOODWARD, OK, USA

WMO: 722152

Lat: 36.437N

Lon: 99.521W

Elev: 2188

StdP: 13.57

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	16.1	-0.1	5.9	21.5	3.0	6.9	24.1	32.0	55.9	28.6	50.7	9.2	0	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	102.4	69.2	99.6	69.5	97.1	69.5	74.7	88.6	73.3	88.0	72.4	87.0	13.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	126.7	79.4	70.1	120.1	78.5	68.4	113.0	78.4	39.7	88.3	38.5	88.4	37.6	86.7	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.2	26.5	24.2	DB	3.1	105.4	5.3	3.7	-0.8	108.0	-3.9	110.2	-6.9	112.2	-10.7	114.9	(4)
			WB	2.0	76.2	4.9	1.3	-1.6	77.2	-4.5	77.9	-7.2	78.7	-10.8	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	38.4	40.0	50.9	58.6	67.9	78.7	83.0	81.2	73.7	60.8	49.0	38.6	(6)
		DBStd	18.50	10.45	11.85	10.73	10.03	9.05	6.18	5.72	6.38	7.12	9.89	10.05	9.79	(7)
		HDD50	1377	378	314	119	34	3	0	0	0	23	136	370		(8)
		HDD65	3808	824	700	448	240	77	1	0	13	198	487	820		(9)
		CDD50	5090	19	34	147	292	557	860	1023	968	712	357	106	15	(10)
		CDD65	2046	0	1	11	47	166	411	558	503	275	67	7	0	(11)
		CDH74	24678	10	43	189	632	1843	4915	7285	6089	2813	748	110	1	(12)
		CDH80	12764	1	10	38	230	785	2540	4204	3396	1275	259	26	0	(13)
(14)	Wind	WSAvg	11.4	10.9	11.0	12.7	13.5	12.0	12.1	10.2	9.8	10.9	11.3	11.5	10.5	(14)
(15)	Precipitation	PrecAvg	24.2	0.7	0.8	1.7	2.0	3.3	3.8	2.4	2.7	1.9	2.1	1.2	1.1	(15)
(16)		PrecMax	30.1	2.8	2.2	5.9	6.9	7.4	7.4	8.5	6.9	4.2	8.7	4.3	4.3	(16)
(17)		PrecMin	17.2	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.6	0.0	0.0	0.1	0.1	(17)
(18)		PrecStd	3.9	0.7	0.6	1.5	1.5	1.8	2.0	1.7	1.5	1.1	2.6	1.3	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	73.4	79.5	84.4	92.7	98.5	104.5	107.7	106.4	99.6	91.2	82.4	70.2		(19)
(20)		MCWB	50.6	51.3	56.6	60.9	66.1	66.9	69.2	68.7	67.2	63.9	56.1	51.0		(20)
(21)		DB	66.4	72.1	79.3	86.6	92.6	99.4	103.6	102.2	94.9	87.6	76.8	63.9		(21)
(22)		MCWB	46.8	50.3	56.5	59.2	65.3	68.1	69.0	70.0	68.0	62.9	54.2	48.0		(22)
(23)		DB	61.1	64.3	73.4	81.3	88.5	95.4	99.7	99.3	90.8	81.9	71.6	59.1		(23)
(24)		MCWB	44.7	48.0	53.7	59.0	64.9	68.6	69.6	70.7	67.5	61.1	53.7	45.0		(24)
(25)		DB	55.0	59.0	69.8	76.6	83.8	91.3	97.1	95.2	88.1	77.4	64.5	54.4		(25)
(26)		MCWB	42.3	45.1	53.3	57.9	64.1	68.7	70.1	70.3	67.0	60.1	51.2	43.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	53.5	57.3	64.3	67.4	71.9	74.9	77.1	76.1	73.5	70.5	61.0	55.2		(27)
(28)		MCDB	68.1	65.2	74.7	77.8	84.6	88.0	90.8	90.4	85.5	82.4	70.2	60.1		(28)
(29)		WB	48.8	52.4	61.3	64.8	70.0	73.1	74.7	74.7	71.5	67.7	58.1	50.3		(29)
(30)		MCDB	65.0	68.0	72.1	76.2	84.4	87.2	88.0	89.3	84.4	78.8	69.3	59.9		(30)
(31)		WB	45.6	49.5	58.4	62.5	68.4	71.9	73.2	73.4	70.2	65.3	55.5	46.8		(31)
(32)		MCDB	60.1	63.7	68.4	73.8	81.9	86.7	88.2	88.5	83.8	76.4	67.4	58.8		(32)
(33)		WB	42.8	46.1	54.5	60.0	66.9	70.8	72.2	72.4	68.9	62.3	53.0	43.8		(33)
(34)		MCDB	54.4	58.0	68.0	72.6	79.2	86.1	87.5	87.0	82.4	73.7	64.2	53.9		(34)
(35)	Mean Daily Temperature Range	MDBR	24.1	24.3	25.4	24.8	23.0	22.9	23.0	23.0	22.9	24.5	25.1	21.6		(35)
(36)		5% DB	32.3	34.2	30.9	31.3	27.9	25.8	25.9	27.3	26.0	29.1	31.8	29.3		(36)
(37)		MCWBR	17.3	17.5	14.1	13.5	10.2	7.5	6.1	6.7	7.9	11.9	14.6	16.3		(37)
(38)		5% WB	31.3	31.4	26.4	24.9	23.5	22.5	21.9	22.7	21.5	23.8	26.3	26.4		(38)
(39)		MCWBR	17.3	17.5	13.8	14.1	10.4	9.1	7.3	7.1	8.1	12.7	14.2	15.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.271	0.280	0.320	0.345	0.360	0.396	0.401	0.389	0.357	0.330	0.291	0.272		(40)
(41)		taud	2.552	2.533	2.429	2.394	2.423	2.363	2.371	2.364	2.432	2.488	2.535	2.554		(41)
(42)		Ebn at Noon	298	307	301	297	292	280	278	280	285	284	287	291		(42)
(43)		Edh at Noon	24	28	34	37	37	39	38	38	34	29	24	23		(43)
(44)	All-Sky Solar Radiation	RadAvg	899	1134	1503	1855	2076	2282	2239	2004	1688	1258	954	782		(44)
(45)		RadStd	81	140	120	123	187	169	115	108	92	164	107	71		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page